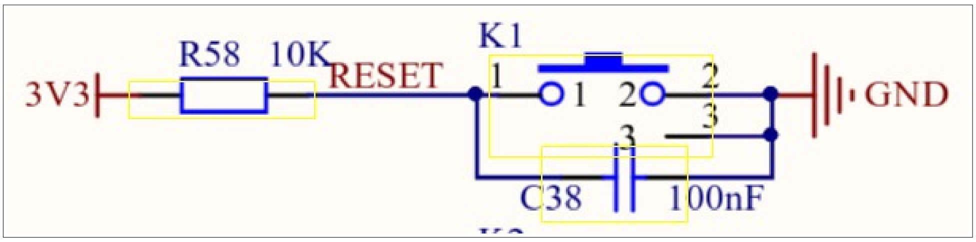
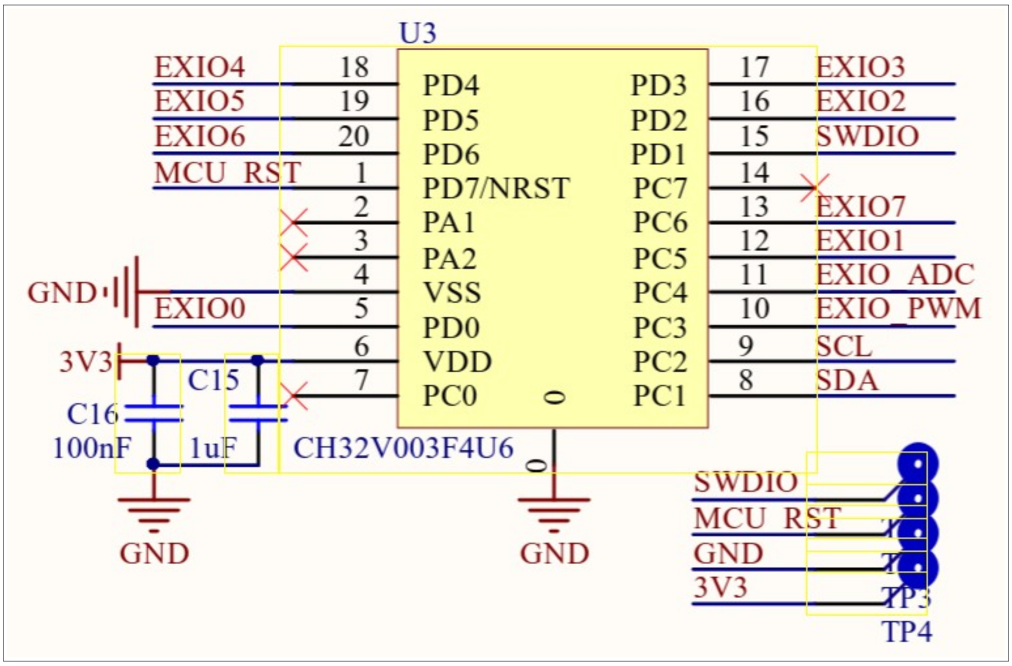
Modding the waveshare board

- **Why ?** The waveshare board rev V4 has a different power circuit for 12 Volt compared to previous versions. As a consequence I noticed that the board regularly failed to start normally when powered from 12 Volt. To start the board a push on the Reset button was necessary.
- **What is the problem ?** A few details on the board could be the cause of this annoying behavior.
  - The RC circuit for the reset of the ESP32S3 I composed of a 10 kΩ resistor and a 100 nF capacitor. This capacitor is 10 times smaller than the capacitor of 1µF Espressif recommends. As a consequence the ESP32S3 is most probably resetted when the power rails are not fully stabilized.



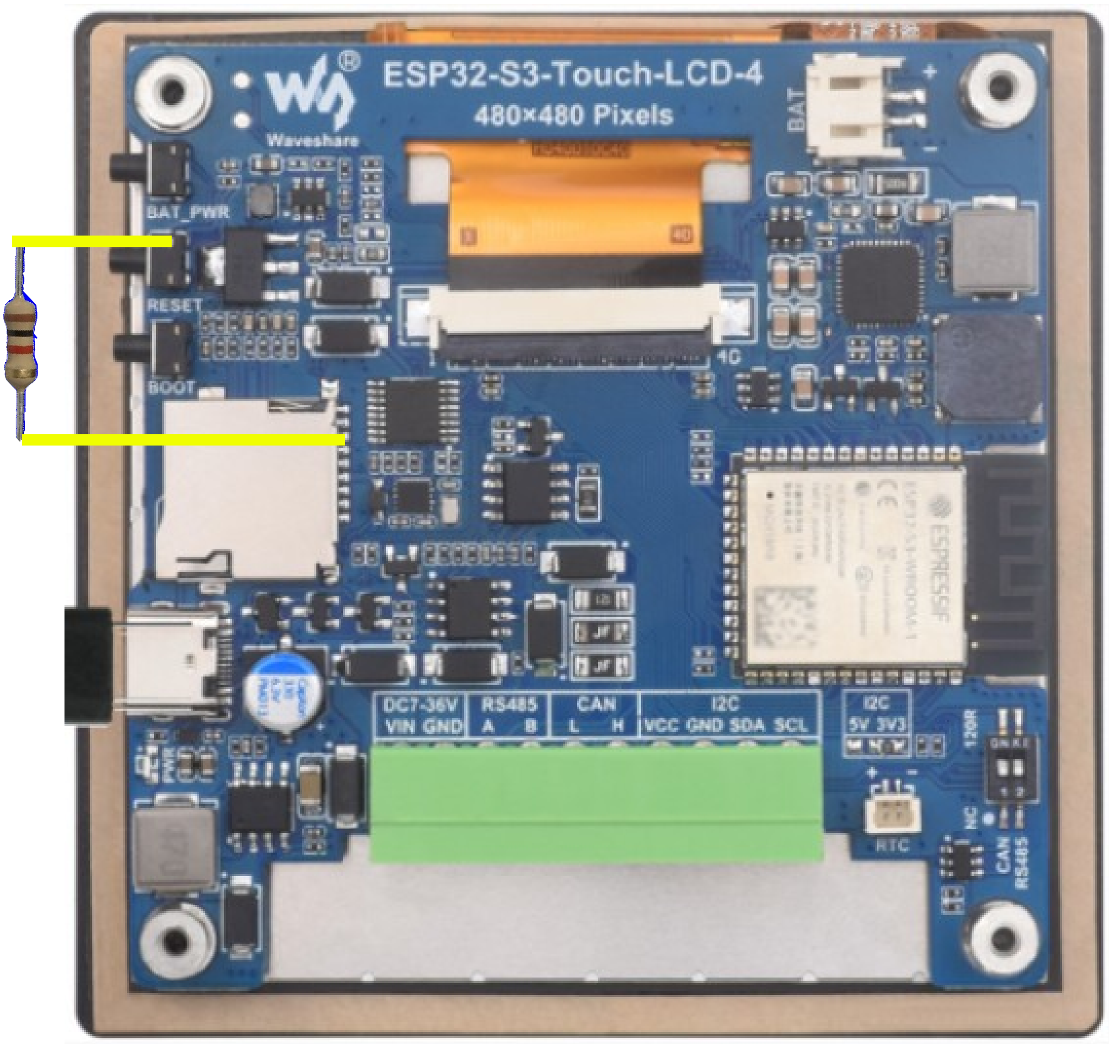
  - The reset pin of the CH32 chip is floating. The CH32 chip is a small microcontroller which is used as an IO expander on this board. By leaving the reset pin floating, the reset timing of the CH32 is uncertain which can cause problems in combination with the first point above.



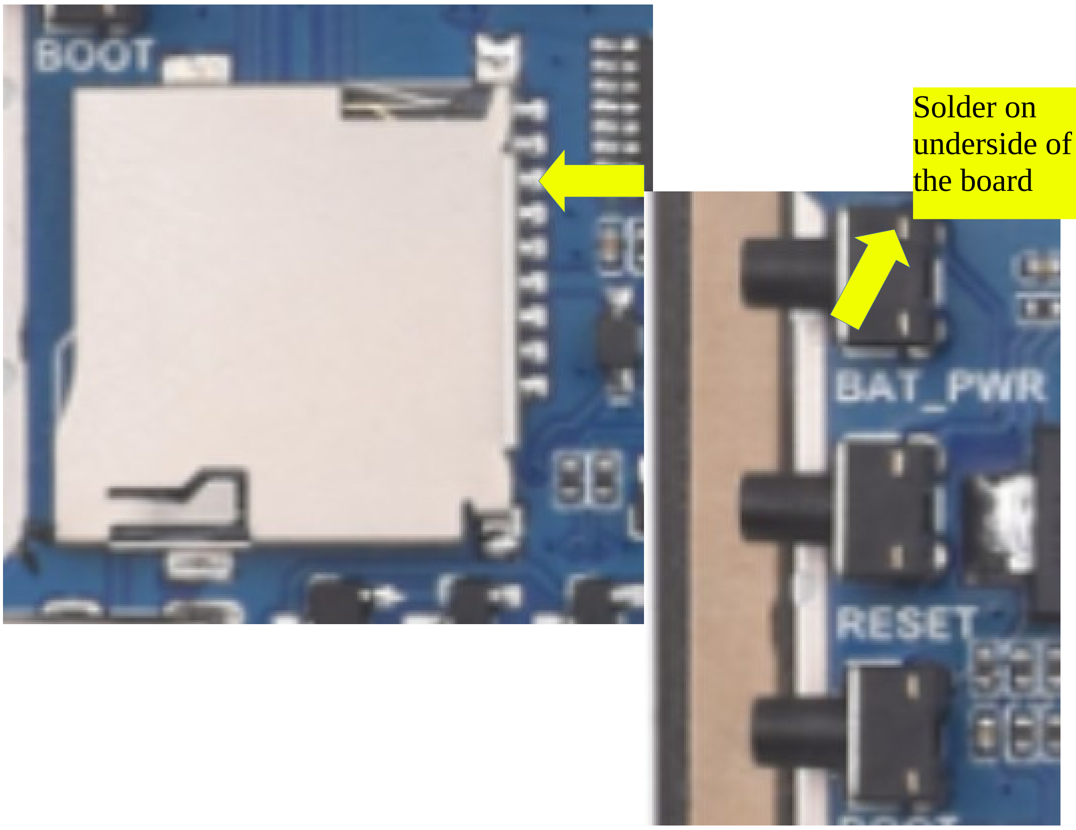
There is test pin2, which can be found under the fpc cable to the display. I tried by forcing a reset on this test pin, but unfortunately this did not solve the cold boot behavior.

- **Solution?**
  - A first possible solution is to replace the capacitor of 100 nF with a 1µF capacitor. The board layout can be found on the wiki for the board: https://files.waveshare.com/wiki/ESP32-S3-Touch-LCD-4/ESP32-S3-Touch-LCD-4%20V4.0.pdf
    I choose not to try and adopt this approach, since I wanted to have control over the reset delay time as to be shure to really solve the problem. (and I hate soldering SMD components.
  - I adopted a different approach: since we are not using the SD card slot for this application, it's IO lines are available. I choose to use IO4 of the SD card slot to provoke a forced hardware reset from this IO line. For this solution to work, 2 modifications are needed:
    - the program has to detect the power up and force 1 extra reset after a delay specified in the program. This small routine has to be placed at the start of Setup in the Arduino sketch. Without this one shot reset modification we would otherwise end up in a reset loop.The board has to be slightly modified. (fortunately no SMD soldering involved). A 1 kΩ resistor has to be placed between IO4 and the reset pin of the ESP32S3, see images below.
      - Soldering the resistor to pin 3 of the SD card slot is straightforward with a steady hand and a soldering iron with a fine tip.

- To solder the other end of the resistor to the reset button the easiest way is to cut through the double sided adhesive foam carefully with an exacto knife. Be cautious not to cut into the flex cable. On my boards the middle foam was not used and did not need to be cut.



- If you scrape the foam a little bit, you have access to the solder pad of the reset button.



- Testing: to test upload the sketch with the mod activated (uncomment line 181 and 184 in setup)

```
176  void setup() {
177      Rev4Board::StartupPrepare();
178      vTaskDelay(100);
179      Beep();
180
181      //pinMode(STARTUP_RESET_GPIO, INPUT);
182      Serial.begin(115200);
183      vTaskDelay(100);
184      //handleOneShotStartupReset();
185
```

- If you now power up the board through USB with your computer, you will notice that the connection with the board is now established at power up and the after a short delay the connection is dropped and the board reconnects. This is the one shot reset doing it’s job.
- After the soldering and testing put some transparent contact glue on both sides of the foam and carefully align the board and display before gluing them together.